
视觉指令微调

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<https://llava-vl.github.io>

Abstract

使用机器生成的指令跟随数据对大型语言模型 (LLMs) 进行指令微调已被证明可以提高其在新任务上的零样本能力,但在多模态领域这一想法尚未得到充分探索。我们提出了首次尝试使用仅语言的 GPT-4 生成多模态语言-图像指令跟随数据。通过对这些生成的数据进行指令微调,我们引入了 LLaVA: Large Language and Vision Assistant, 这是一个端到端训练的大规模多模态模型,它将视觉编码器和 LLM 连接起来以实现通用的视觉和语言理解。为了促进未来对视觉指令跟随的研究,我们构建了两个具有多样化和挑战性应用导向任务的评估基准。我们的实验表明,LLaVA 展示了令人印象深刻的多模态聊天能力,在某些情况下,在未见过的图像/指令上表现出多模态 GPT-4 的行为,并在合成多模态指令跟随数据集上获得了相对于 GPT-4 的 85.1% 相对得分。当在 Science QA 上进行微调时,LLaVA 与 GPT-4 的协同作用达到了新的 92.53% 的准确率。我们将 GPT-4 生成的视觉指令调优数据、我们的模型和代码公开提供。

1 引言

人类通过许多通道(如视觉和语言)与世界互动,因为每个单独的通道在表示和传达某些概念方面具有独特的优势,从而有助于更好地理解世界。人工智能的一个核心目标是开发一种通用助手,能够有效遵循多模态视觉-语言指令,与人类意图保持一致,以完成野外的各种现实任务 [4, 27, 26]。

为此,社区见证了一种新兴的兴趣,即开发语言增强的基础视觉模型 [27, 16], 这些模型在开放世界视觉理解方面具有强大的能力,例如分类 [40, 21, 57, 54, 39]、检测 [29, 62, 33]、分割 [25, 63, 58] 和标注 [50, 28], 以及视觉生成和编辑 [42, 43, 56, 15, 44, 30]。我们建议读者查阅《野外计算机视觉》阅读列表以获取更及时的文献综述 [12]。在这类工作中,每个任务都由一个单独的大规模视觉模型独立解决,任务指令隐式地包含在模型设计中。此外,语言仅用于描述图像内容。虽然这允许语言在将视觉信号映射到语言语义方面发挥重要作用——这是人类交流的通用渠道,但它导致了模型通常具有固定的接口,与用户指令的交互性和适应性有限。

大型语言模型 (LLM) 另一方面已经表明,语言可以发挥更广泛的作用:通用接口为通用助手,其中各种任务指令可以在语言中显式表示,并引导端到端训练的神经助手切换到感兴趣的任務以解决问题。例如,ChatGPT [35] 和 GPT-4 [36] 的最新成功展示了对齐的 LLM 在

遵循人类指令方面的强大能力，并激发了开发开源 LLM 的极大兴趣。其中，LLaMA [49] 是一个开源的 LLM，其性能可与 GPT-3 匹敌。Alpaca [48]、Vicuna [9]、GPT-4-LLM [38] 利用各种机器生成的高质量指令跟随样本来提高 LLM 的对齐能力，与专有 LLM 相比表现出令人印象深刻的结果。重要的是，这项工作 是 纯文本 的。

在本文中，我们提出了视觉指令微调，这是首次尝试将指令微调扩展到语言-图像多模态空间，以迈向构建通用视觉助手。特别是，我们的论文做出了以下贡献：

- 多模态指令跟随数据。一个关键挑战是缺乏视觉-语言指令跟随数据。我们提出了一个数据重构视角和流水线，使用 ChatGPT/GPT-4 将图像-文本对转换为适当的指令跟随格式。
- 大型多模态模型。我们通过将 CLIP [40] 的开放集视觉编码器与语言解码器 Vicuna [9] 连接起来，并在我们生成的指令视觉-语言数据上进行端到端微调，开发了一个大型多模态模型 (LMM)。我们的实证研究表明，使用生成的数据对 LMM 进行指令微调是有效的，并为构建通用指令跟随视觉智能体提供了实用建议。当与 GPT-4 集成时，我们的方法在 Science QA [34] 多模态推理数据集上达到了最新技术水平。
- 多模态指令跟随基准。我们提出了 LLaVA-Bench，它包含两个具有挑战性的基准，具有多样化的图像、指令和详细标注的配对数据。
- 开源。我们向公众发布了以下资产：生成的多模态指令数据、代码库、模型检查点以及一个视觉聊天演示。

2 相关工作

多模态指令跟随智能体。在计算机视觉中，现有的构建指令跟随智能体的工作可以大致分为两类：(i) 端到端训练的模型，这些模型是为每个特定的研究主题分别探索的。例如，视觉-语言导航任务 [3, 19] 和 Habitat [47] 要求具身人工智能代理遵循自然语言指令，并采取一系列动作在视觉环境中完成目标。在图像编辑领域，给定一个输入图像和一个书面指令告诉代理做什么，InstructPix2Pix [6] 通过遵循人类指令来编辑图像。(ii) 一种通过 LangChain [1] / 大型语言模型 (LLMs) [35] 协调各种模型的系统，例如 Visual ChatGPT [53]、X-GPT [63]、MM-REACT [55]、VisProg [18] 和 ViperGPT [46]。虽然我们与这些工作有着相同的指令跟随智能体构建目标，但我们专注于开发一个端到端训练的语言-视觉多模态模型以用于 多个 任务。

指令微调。在自然语言处理 (NLP) 社区中，为了使像 GPT-3 [7]、T5 [41]、PaLM [10] 和 OPT [60] 这样的大型语言模型 (LLMs) 能够遵循自然语言指令并完成现实世界中的任务，研究人员探索了 LLMs 指令微调的方法 [37, 52, 51]，从而产生了如 InstructGPT [37]/ChatGPT [35]、FLAN-T5 [11]、FLAN-PaLM [11] 和 OPT-IML [22] 等指令微调的对应模型。结果表明，这种方法可以有效提高 LLMs 的零样本和少样本泛化能力。因此，借用 NLP 的想法到计算机视觉是自然而然的。更广泛地说，在诸如图像分类 [14] 等其他主题中也研究了基于基础模型的教师-学生蒸馏思想。Flamingo [2] 可以被视为多模态领域的 GPT-3 时刻，由于其在零样本任务迁移和上下文学习方面的强大性能。其他在图像-文本对上训练的多模态语言模型 (LMMs) 包括 BLIP-2 [28]、FROMAGe [24] 和 KOSMOS-1 [20]。PaLM-E [13] 是用于具身 AI 的 LMM。基于最近的“最佳”开源 LLM LLaMA，OpenFlamingo [5] 和 LLaMA-Adapter [59] 是开源的努力，使 LLaMA 能够使用图像输入，为构建开源多模态 LLMs 铺平了道路。尽管这些模型展示了有希望的任务迁移泛化性能，但它们并非明确地用视觉-语言指令数据进行微调，并且它们在多模态任务中的表现通常比仅语言任务差。本文旨在填补这一空白并研究其有效性。最后，请注意视觉指令微调与视觉提示微调 [23] 不同：前者旨在提高模型的指令跟随能力，而后者旨在提高模型适应过程中的参数效率。

Context type 1: Captions A group of people standing outside of a black vehicle with various luggage surrounds a vehicle in an underground parking area People try to fit all of their luggage in an SUV. The sport utility vehicle is parked in the public garage, being packed for a trip. Some people with luggage near a van that is transporting it. Context type 2: Boxes person: [0.681, 0.242, 0.774, 0.694], backpack: [0.384, 0.696, 0.485, 0.914], suitcase: ...<omitted>	
Response type 1: conversation Question: What type of vehicle is featured in the image? Answer: The image features a black sport utility vehicle (SUV) ...<omitted> Response type 2: detailed description The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip. ...<omitted> Response type 3: complex reasoning Question: What challenges do these people face? Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings ...<omitted>	

表 1: 一个说明指令跟随数据的示例。顶部的块显示用于提示 GPT 的上下文，例如标题和框，底部的块显示三种类型的响应。请注意，视觉图像未用于提示 GPT，我们仅在此处将其作为参考展示。

3 GPT 辅助的视觉指令数据生成

社区见证了大量公开的多模态数据（如图像-文本对）的增长，范围从 CC [8] 到 LAION [45]。然而，当涉及到多模态指令跟随数据时，可用的数量是有限的，部分原因是创建此类数据的过程耗时且不够明确，特别是当考虑人工众包时。受最近 GPT 模型在文本注释任务中成功 [17] 的启发，我们提议利用 ChatGPT/GPT-4 基于广泛存在的图像对数据进行多模态指令跟随数据收集。

对于一幅图像 X_v 及其关联的标题 X_c ，自然会生成一组问题 X_q ，目的是指导助手描述图像内容。我们提示 GPT-4 策划了这样一份问题列表（详见附录）。因此，将图像-文本对扩展为指令跟随版本的一种简单方法是 **Human** : X_q X_v <STOP> **Assistant** : X_c <STOP>。虽然构建成本低廉，但这种简单的扩展版本在指令和响应中都缺乏多样性和深入推理。

为了解决这个问题，我们利用仅接受文本输入的语言型 GPT-4 或 ChatGPT 作为强大的教师模型，来创建涉及视觉内容的指令跟随数据。具体来说，为了将图像编码为其视觉特征以提示仅支持文本的 GPT，我们使用两种类型的符号表示：(i) 描述语通常从不同角度描述视觉场景；(ii) 边界框通常用于定位场景中的对象，每个框编码对象的概念及其空间位置。一个样例显示在表 14 的顶部块中。

这种符号表示法使我们能够将图像编码为 LLM 可识别的序列。我们使用 COCO 图像 [31]，并生成三种类型的指令跟随数据。表 14 底部块中每种类型显示一个示例。对于每种类型，我们首先手动设计一些示例。它们是我们数据收集过程中唯一的标注，在提示学习中用作种子示例来查询 GPT-4。

- 对话. 我们设计了一个助理与提问者之间的对话，提问者针对这张照片提出问题。答案的语气就像是助理正在看图像并回答问题一样。提出了一系列关于图像视觉内容的问题，包括物体类型、物体计数、物体 Action、物体位置、物体之间的相对位置等问题。只考虑有明确答案的问题。有关详细提示，请参见附录。
- 详细描述. 为了包含对图像的丰富且全面的描述，我们创建了一份旨在实现此目标的问题列表。然后我们提示 GPT-4 并整理该列表（有关提示和整理过程的详细信息，请参见附录）。对于每张图像，我们从列表中随机抽取一个问题，让 GPT-4 生成详细的描述。
- 复杂推理. 上述两种类型侧重于视觉内容本身，基于此我们进一步创建深入推理问题。答案通常需要遵循严密的逻辑逐步推理得出。

我们总共收集了 158K 个独特的语言-图像指令跟随样本，其中包括会话中的 58K 个、详细描述中的 23K 个和复杂推理中的 77K 个。我们在早期实验中测试了 ChatGPT 和 GPT-4 的使用，发现 GPT-4 始终提供更高质量的指令跟随数据，例如空间推理。

4 视觉指令微调

4.1 架构

主要目标是有效利用预训练语言模型 (LLM) 和视觉模型的能力。网络架构如图 1 所示。我们选择 Vicuna[9] 作为我们由 ϕ 参数化的语言模型 $f_\phi(\cdot)$ ，因为在公开可用的检查点中，它在语言任务中具有最佳的指令跟随能力 [48, 9, 38]。

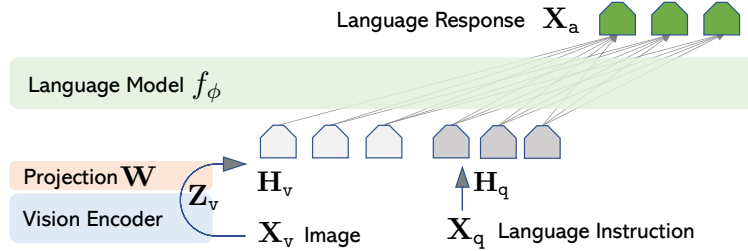


图 1: LLaVA 网络结构.

对于输入图像 X_v ，我们考虑预训练的 CLIP 视觉编码器 ViT-L/14 [40]，它提供了视觉特征 $Z_v = g(X_v)$ 。在最后一个 Transformer 层之前和之后的网格特征在我们的实验中被考虑。我们考虑一个简单的线性层以将图像特征连接到词嵌入空间。具体来说，我们应用一个可训练的投影矩阵 W 将 Z_v 转换为语言嵌入词元 H_v ，它们与语言模型中的词嵌入空间具有相同的维度：

$$H_v = W \cdot Z_v, \text{ with } Z_v = g(X_v) \quad (1)$$

因此，我们有一系列视觉词元 H_v 。请注意，我们的简单投影方案是轻量级的，这使我们可以快速迭代以数据为中心的实验。还可以考虑更复杂的方案来连接图像和语言表示，例如 Flamingo [2] 中的门控交叉注意力和 BLIP-2 [28] 中的 Q-former。我们将探索可能更有效且更复杂的 LLaVA 架构设计作为未来的工作。


```

Xsystem-message <STOP>
Human : Xinstruct1 <STOP> Assistant: Xa1 <STOP>
Human : Xinstruct2 <STOP> Assistant: Xa2 <STOP> ...

```

表 2: 输入序列用于训练模型。这里仅展示了两个对话轮次；实际上，轮次数根据指令跟随数据而变化。在我们目前的实现中，我们遵循 Vicuna-v0 [9] 设置系统消息 $X_{\text{system-message}}$ ，并将<STOP>设置为###。模型被训练以预测助手的回答以及何时停止，因此仅绿色序列/词元用于自动回归模型中的损失计算。

4.2 训练

对于每张图像 X_v ，我们生成多轮对话数据 $(X_q^1, X_a^1, \dots, X_q^T, X_a^T)$ ，其中 T 是总轮数。我们将它们组织成一个序列，将所有答案视为助手的回复，并在第 t 轮的指令 X_{instruct}^t 表示为：

$$X_{\text{instruct}}^t = \begin{cases} \text{Randomly choose } [X_q^1, X_v] \text{ or } [X_v, X_q^1], & \text{the first turn } t = 1 \\ X_q^t, & \text{the remaining turns } t > 1 \end{cases} \quad (2)$$

这导致了在表 2 中说明的多模态指令跟随序列的统一格式。我们使用原始的自回归训练目标，在预测词元上对 LLM 进行指令调整。

具体地，对于长度为 L 的序列，我们通过以下方式计算目标答案的概率 X_a ：

$$p(X_a | X_v, X_{\text{instruct}}) = \prod_{i=1}^L p(\mathbf{x}_i | X_v, X_{\text{instruct}, <i}, X_{a, <i}), \quad (3)$$

其中 θ 是可训练的参数， $X_{\text{instruct}, <i}$ 和 $X_{a, <i}$ 分别是当前预测词元 $\mathbf{x}_i$ 之前所有轮次中的指令和答案词元。请参见表 2 了解预测词元的示例。对于 (3) 中的条件部分，我们显式添加 X_v 来强调图像在所有答案中都被关联的事实，并且我们省略 $X_{\text{system-message}}$ 和之前所有的 <STOP> 以提高可读性。对于 LLaVA 模型训练，我们采用两阶段指令微调过程。

阶段 1: 特征对齐的预训练。 为在概念覆盖和训练效率之间取得平衡，我们将 CC3M 过滤为 595K 图像-文本对。请参见附录以了解过滤过程的详细信息。这些对使用第 3 节中描述的简单扩展方法转换为指令跟随数据。每个样本可以被视为一次单轮对话。为了构建 (2) 中的输入 X_{instruct} ，对于一张图像 X_v ，随机采样一个问题 X_q ，这是一个语言指令，要求助手简要描述图像。真实答案 X_a 是原始标题。在训练过程中，我们保持视觉编码器和 LLM 权重冻结，并仅最大化 (3) 的似然性，其可训练参数为 $\theta = W$ (投影矩阵)。这样，图像特征 H_v 可以与预训练的 LLM 词嵌入对齐。此阶段可以理解为为冻结的 LLM 训练一个兼容的视觉词元分析器。

阶段 2: 端到端微调。 我们始终保持视觉编码器权重冻结，并继续更新 LLaVA 中投影层和 LLM 的预训练权重；即，可训练参数为 $\theta = \{W, \phi\}$ ((3) 中的)。我们考虑两种具体的使用场景：

- 多模态聊天机器人。我们通过在第 3 节的 158K 语言-图像指令跟随数据上进行微调来开发一个聊天机器人。在这三种类型的响应中，对话是多轮的，而其他两种是单轮的。它们在训练过程中均匀采样。
- Science QA。我们在 ScienceQA 基准 [34] 上研究了我们的方法，这是第一个大规模的多模态科学问题数据集，其中答案带有详细的课程和解释。每个问题都以自然语言或图像的形式提供上下文。助手以自然语言提供推理过程，并从多个选项中选择答案。对于

Visual input example, Extreme Ironing:



Source: <https://www.barnorama.com/wp-content/uploads/2016/12/03-Confusing-Pictures.jpg>

User	What is unusual about this image?
LLaVA	The unusual aspect of this image is a man ironing clothes on the back of a minivan or van. This is not a typical place to perform this activity, as one would usually iron clothes in a more stationary and safe location, such as a home, using a regular ironing board. The scene depicted in the image is peculiar as it involves a makeshift ironing setup on a vehicle, which can be both unsafe and unconventional. Additionally, it is not clear how the man is able to maintain balance and stability while ironing clothes in such an unstable environment. [Start a new conversation, and clear the history]
User	What' s happening in the scene?
LLaVA	The image depicts a man standing on top of a yellow SUV in a busy city street. He is holding a portable ladder, seemingly ironing clothes while standing on the vehicle. Around the scene, there are other cars, a traffic light, and a fire hydrant. Additionally, there are two other people in the vicinity, one close to the man on the car and another further away. The man on the ladder appears to be attracting attention due to his unconventional choice of ironing his clothes on top of a moving car . The city street around him is bustling with activity, adding to the unique nature of the scene.
User	What is unusual about this image?
GPT-4 [36]	The unusual thing about this image is that a man is ironing clothes on an ironing board attached to the roof of a moving taxi.
User	What is unusual about this image?
BLIP-2	a man is sitting on the back of a yellow cab
User	What is unusual about this image?
OpenFlamingo	The man is drying his clothes on the hood of his car.

表 3: Example prompt from GPT-4 paper [36] to compare visual reasoning and chat capabilities. Compared to BLIP-2 [28] and OpenFlamingo [5], LLaVA accurately follows the user’s instructions, instead of simply describing the scene. LLaVA offers a more comprehensive response than GPT-4. Even when merely asked to describe the image, LLaVA identifies atypical aspects of the image.

在 (2) 中的训练, 我们将数据组织为单轮对话, 问题&上下文作为 X_{instruct} , 推理&答案作为 X_a 。

5 实验

我们通过两种主要的实验情景: 多模态聊天机器人和 ScienceQA 数据集, 分别评估了 LLaVA 在指令跟随和视觉推理能力方面的表现。我们按照 Vicuna 的超参数 [9] 使用 $8\times$ 个 A100 显

	Conversation	Detail description	Complex reasoning	All
Full data	83.1	75.3	96.5	85.1
Detail + Complex	81.5 (-1.6)	73.3 (-2.0)	90.8 (-5.7)	81.9 (-3.2)
Conv + 5% Detail + 10% Complex	81.0 (-2.1)	68.4 (-7.1)	91.5 (-5.0)	80.5 (-4.4)
Conversation	76.5 (-6.6)	59.8 (-16.2)	84.9 (-12.4)	73.8 (-11.3)
No Instruction Tuning	22.0 (-61.1)	24.0 (-51.3)	18.5 (-78.0)	21.5 (-63.6)

表 4: 在 LLaVA-Bench (COCO) 上进行消融实验, 使用不同的训练数据。我们报告相对于仅使用文本的 GPT-4 模型的相对得分, 该模型以真实值图像描述和边界框作为视觉输入。我们将 GPT-4 提示为我们的模型输出的答案以及 GPT-4 (仅文本) 的答案, 并让它比较两者的响应并给出评分及解释。

卡训练所有模型。我们在过滤后的 CC-595K 子集上对模型进行 1 轮预训练, 学习率为 $2e-3$, 批量大小为 128, 并在提出的 LLaVA-Instruct-158K 数据集上进行 3 轮微调, 学习率为 $2e-5$, 批量大小为 32。有关更多训练细节, 请参见附录。

5.1 多模态聊天机器人

我们开发了一个聊天机器人演示程序, 以展示 LLaVA 的图像理解和对话能力, 并研究 LLaVA 消化视觉输入并表现出指令跟随能力的程度。我们首先使用 GPT-4 原始论文 [36] 中的样例 (如表 3 所示, 更多样例见附录), 这些样例需要深入的图像理解。为了比较, 我们引用了多模态 GPT-4 论文中的查询和响应, 并查询 BLIP-2 和 OpenFlamingo 模型检查点以获取它们的响应。

令人惊讶的是, 尽管 LLaVA 是在一个小的多模态指令跟随数据集 ($\sim 80K$ 独特图像) 上训练的, 它在这些样例上的推理结果与多模态 GPT-4 非常相似。请注意, 虽然这些图像是 LLaVA 的领域外数据, LLaVA 仍然能够理解场景并遵循问题指令提供合理的回答。相比之下, BLIP-2 和 OpenFlamingo 专注于描述图像, 而不是按照用户指令以适当的方式作答。

定量评估。 为了系统地了解 LLaVA 的表现, 我们提出了一种定量指标来衡量模型在多模态数据上的指令跟随能力。受 [9] 的启发, 我们利用 GPT-4 来衡量生成响应的质量。具体来说, 我们创建由图像、真实文本描述和问题组成的三元组。候选模型 (例如 LLaVA) 根据问题和图像预测答案。为了提供一个近似的理论上限, 我们基于问题和真实值文本描述, 使用仅文本的 GPT-4 创建一个参考预测。在获得两个模型的响应后, 我们将问题、视觉信息 (以文本描述的形式)、以及来自两个助手的生成响应传递给裁判 (即仅文本的 GPT-4)。它评估助手响应的有用性、相关性、准确性以及详细程度, 并在 1 到 10 的评分尺度上给出总体分数, 分数越高表示整体表现越好。它也被要求提供全面的解释, 以便我们更好地理解模型。我们将相对分数报告为相对于使用文本真实值描述作为视觉输入的仅文本 GPT-4 模型的结果。我们创建了两个基准来评估模型的性能。

LLaVA-Bench (COCO). 我们从 COCO-Val-2014 中随机选择 30 张图像, 并使用第 3 节提出的流水线为每张图像生成三种类型的问题 (对话、详细描述、复杂推理), 总共生成 90 个问题。该基准研究了模型在一致视觉输入下的对齐行为和能力。我们改变训练数据集以研究不同类型指令跟随数据的有效性, 并在表 4 中展示结果。首先, 通过指令调整, 模型的指令跟随能力显著提高了 50 分以上。其次, 添加少量详细描述和复杂推理问题有助于模型整体能力提高 7 分。此外, 它还改善了模型在对话问题上的表现, 表明推理能力的提升可以补充对话能力。最后, 我们展示了拥有三种类型数据的表现最佳, 达到了 85.1%。

	Conversation	Detail description	Complex reasoning	All
OpenFlamingo [5]	19.3 $\pm$ 0.5	19.0 $\pm$ 0.5	19.1 $\pm$ 0.7	19.1 $\pm$ 0.4
BLIP-2 [28]	54.6 $\pm$ 1.4	29.1 $\pm$ 1.2	32.9 $\pm$ 0.7	38.1 $\pm$ 1.0
LLaVA	57.3 $\pm$ 1.9	52.5 $\pm$ 6.3	81.7 $\pm$ 1.8	67.3 $\pm$ 2.0
LLaVA [†]	58.8 $\pm$ 0.6	49.2 $\pm$ 0.8	81.4 $\pm$ 0.3	66.7 $\pm$ 0.3

表 5: 使用 LLaVA-Bench (In-the-Wild) 上的相对得分进行指令跟随能力比较。结果以均值 $\pm$ 标准差的格式报告。对于前三行, 我们报告三次推理运行。LLaVA 的表现明显优于其他方法。[†] 对于给定的一组 LLaVA 解码序列, 我们通过向 GPT-4 查询三次来进行评估; GPT-4 给出了一致的评估。

LLaVA-Bench (In-the-Wild). 为了评估模型在更具挑战性的任务中的能力和对新领域的泛化能力, 我们收集了一组包含 24 张图像和总共 60 个问题的多样化数据集, 其中包括室内和室外场景、表情包、绘画、素描等, 并为每张图像提供高度详细且手动精心编写的描述以及一组合适的问题。我们在表 5 中比较了 LLaVA、BLIP 和 OpenFlamingo。得益于视觉指令微调, LLaVA 相比 BLIP-2 (+29%) 和 OpenFlamingo (+48%) 取得了显著更好的性能。与可以访问真实标签的纯文本 GPT-4 相比, LLaVA 在复杂推理问题上取得了令人印象深刻的 81.7% 的性能, 在整体得分为 67.3% 的情况下表现优异。

局限性。 这个 LLaVA-Bench (In-the-Wild) 旨在具有挑战性并揭示模型的弱点。我们在表 6 中提供了两个带有相关标题和问题的例子。对于拉面例子 (左边), 要正确回答餐厅名称, 模型需要具备广泛的知识覆盖范围和多语言理解能力; 要正确描述配菜, 模型可能需要从互联网检索相关的多模态信息。对于冰箱例子 (右边), 识别酸奶的正确品牌需要模型处理高分辨率图像并具备广泛的知识覆盖范围。我们还观察到一个有趣的失败案例: 当被问及是否含有草莓味酸奶时, LLaVA 回答了“是”, 尽管冰箱里只有酸奶和草莓。这表明有时 LLaVA 将图像视为“补丁的集合”, 未能理解图像中的复杂语义。我们希望 LLaVA 能作为基准测试的坚实基线, 在其上我们的发现能够启发未来开发更强大的 LMM 的工作。

5.2 科学问答

ScienceQA [34] 包含 21k 多模态选择题, 具有丰富的领域多样性, 涵盖 3 个学科、26 个主题、127 个类别和 379 项技能。基准数据集被划分为训练、验证和测试集, 分别包含 12726、4241 和 4241 个样例。我们考虑了两种代表性方法, 包括带有和不带思维链 (CoT) 的 GPT-3.5 模型 (`text-davinci-002`), LLaMA-Adapter [59], 以及多模态思维链 (MM-CoT) [61], 这是该数据集当前最先进的方法。更多基准结果, 请参见 [34]。

结果见表 7。对于 LLaVA, 我们使用倒数第二层之前的视觉特征, 要求模型先预测原因再预测答案, 并训练 12 个轮次。其准确率达到 90.92%, 与最先进的技术 (SoTA) 91.68% 非常接近。为了探索大型语言模型 (LLMs) 的极限, 我们还使用了 2-shot 上下文学习来提示 GPT-4, 并取得了 82.69% 的准确率, 比 GPT-3.5 的 75.17% 绝对提高了 7.52%。在大量问题中, 我们注意到 GPT-4 失败的原因很简单, 因为它报告说缺乏上下文信息, 例如图像或图表。我们考虑了两种方案来结合我们的模型和 GPT-4 的结果。(i) GPT-4 补充。每当 GPT-4 无法提供答案时, 我们就使用我们的方法进行预测。此方案达到了 90.97% 的准确率, 几乎与单独应用我们的方法相同。(ii) GPT-4 作为裁判。每当 GPT-4 和 LLaVA 产生不同的答案时, 我们再次提示 GPT-4, 要求它基于问题和两个结果给出自己的最终答案。这种方法的精神类似于 CoT, 但具有来自其他模型的外部知识。令人惊讶的是, 此方案能够在所有问题类别上提供一致的改进, 并实现了新的 SoTA 准确率为 92.53%。有趣的是, 不能处理图像的纯文本 GPT-4, 在那些以图像为上下文的问题上, 提高了模型的整体性能。这是因为其中一

Challenging examples from LLaVA-Bench (In-the-Wild):



ICHIRAN Ramen [source]



Filled fridge [source]

Annotation	A close-up photo of a meal at ICHIRAN. The chashu ramen bowl with a spoon is placed in the center. The ramen is seasoned with chili sauce, chopped scallions, and served with two pieces of chashu. Chopsticks are placed to the right of the bowl, still in their paper wrap, not yet opened. The ramen is also served with nori on the left. On top, from left to right, the following sides are served: a bowl of orange spice (possibly garlic sauce), a plate of smoke-flavored stewed pork with chopped scallions, and a cup of matcha green tea.	An open refrigerator filled with a variety of food items. In the left part of the compartment, towards the front, there is a plastic box of strawberries with a small bag of baby carrots on top. Towards the back, there is a stack of sauce containers. In the middle part of the compartment, towards the front, there is a green plastic box, and there is an unidentified plastic bag placed on it. Towards the back, there is a carton of milk. In the right part of the compartment, towards the front, there is a box of blueberries with three yogurts stacked on top. The large bottle of yogurt is Fage non-fat yogurt, and one of the smaller cups is Fage blueberry yogurt. The brand and flavor of the other smaller cup are unknown. Towards the back, there is a container with an unknown content.
Question 1	What's the name of the restaurant?	What is the brand of the blueberry-flavored yogurt?
Question 2	Describe this photo in detail.	Is there strawberry-flavored yogurt in the fridge?

表 6: Challenging examples from LLaVA-Bench (In-the-Wild), we provide extremely-detailed annotation for each image for an accurate evaluation. Some questions require the model to extract details from high resolution image and to have a broad knowledge coverage.

Method	Subject			Context Modality			Grade		Average
	NAT	SOC	LAN	TXT	IMG	NO	G1-6	G7-12	
Representative & SoTA methods with numbers reported in the literature									
Human [34]	90.23	84.97	87.48	89.60	87.50	88.10	91.59	82.42	88.40
GPT-3.5 [34]	74.64	69.74	76.00	74.44	67.28	77.42	76.80	68.89	73.97
GPT-3.5 w/ CoT [34]	75.44	70.87	78.09	74.68	67.43	79.93	78.23	69.68	75.17
LLaMA-Adapter [59]	84.37	88.30	84.36	83.72	80.32	86.90	85.83	84.05	85.19
MM-CoT _{Base} [61]	87.52	77.17	85.82	87.88	82.90	86.83	84.65	85.37	84.91
MM-CoT _{Large} [61]	95.91	82.00	90.82	95.26	88.80	92.89	92.44	90.31	91.68
Results with our own experiment runs									
GPT-4 [†]	84.06	73.45	87.36	81.87	70.75	90.73	84.69	79.10	82.69
LLaVA	90.36	95.95	88.00	89.49	88.00	90.66	90.93	90.90	90.92
LLaVA+GPT-4 [†] (complement)	90.36	95.50	88.55	89.05	87.80	91.08	92.22	88.73	90.97
LLaVA+GPT-4 [†] (judge)	91.56	96.74	91.09	90.62	88.99	93.52	92.73	92.16	92.53

表 7: 准确率 (%) 在 Science QA 数据集上。问题类别: NAT = 自然科学, SOC = 社会科学, LAN = 语言科学, TXT = 文本上下文, IMG = 图像上下文, NO = 无上下文, G1-6 = 1-6 年级, G7-12 = 7-12 年级。[†] 仅文本的 GPT-4, 我们的评估。我们提出的模型集成方法始终在所有类别中提升模型性能, 达到新的 SOTA 表现。

些问题实际上不需要图像上下文就能得到正确答案。GPT-4 裁判可以识别这些情况并纠正 LLaVA 的一些错误。详见附录。据我们所知, 这是首次将 GPT-4 用于模型集成。我们希望这一发现能够鼓励未来的研究探索更多有效的利用大型语言模型进行模型集成的方法。

消融实验。 我们在表 8 中对 ScienceQA 的几个设计选择进行了消融实验。(i) 视觉特征。我们尝试使用 CLIP 视觉编码器的最后一层特征, 这得到了 89.96% 的准确率, 比倒数第二层的特征低 0.96%。我们假设这是因为与倒数第二层相比, CLIP 的最后一层特征可能更多地关注全局和抽象的图像属性, 而倒数第二层可以更关注于理解特定图像细节有用的局部属性。(ii) 思维链。为了决定模型预测中答案和推理过程的顺序, 我们运行了两种变体并观察到, 在 12 轮次中, 答案优先报告了最好的 89.77% 准确率, 而推理优先可以在 6 轮次内快速达到 89.77% 的准确率, 但进一步训练不会带来更多的提升。训练模型 24 轮次并未提高性能。我们得出结论, 类似于 CoT 的推理优先策略可以大大加快收敛速度, 但对最终性能的贡献相对较小。(iii) 预训练。我们跳过预训练, 直接从零开始在 Science QA 上训练 – 准确率下降到 85.81%。5.11% 的绝对下降表明了我们的预训练阶段的重要性, 它在对齐多模态特征的同时保留了大量的预训练知识。(iv) 模型大小。我们将所有配置保持为最佳的 13B 模型, 并训练一个 7B 模型。这得到了 89.84% 的准确率, 比 90.92% 低 1.08%, 这证明了模型规模的重要性。

Visual features	Before	Last
Best variant	90.92	89.96 (-0.96)
Predict answer first	-	89.77 (-1.15)
Training from scratch	85.81 (-5.11)	-
7B model size	89.84 (-1.08)	-

表 8: Design choice ablations (%). The difference with the best variant is reported in red text.

6 结论

本文展示了视觉指令微调的有效性。我们提出了一种自动化的流水线来创建语言-图像指令跟随数据, 基于此我们训练了 LLaVA, 一个多模态模型, 以遵循人类意图完成视觉任务。当

在 ScienceQA 上进行微调时，它达到了新的 SoTA 准确率，并且在多模态聊天数据上微调时具有出色的视觉聊天能力。此外，我们提出了第一个基准来研究多模态指令跟随能力。本文是视觉指令微调的初步步骤，主要关注现实任务。有关 LLaVA 在学术基准上的更多定量结果，请参考带有视觉指令微调的改进基线 [32]。我们希望我们的工作能启发未来构建更强大的多模态模型的研究。

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A 更广泛的影响

通用视觉助手 LLaVA 的更广泛影响在其部署和发布方面存在潜在的益处和风险。由于其视觉特性，一些考量因素是 LLaVA 所独有的，而其他一些则与现有的遵循指令的语言模型（例如 Alpaca、Vicuna 等）有相似之处。由于 LLaVA 是基于 LLaMA、Vicuna 和 CLIP 构建的，它继承了一些与语言模型和视觉编码器相关的问题。在下面，我们概述了该模型发布时的风险和缓解策略。

恶意输入。 为尽量减少潜在的误用和有害后果，我们为 LLaVA 采取了两种预防措施：(1) 使用 OpenAI Filter API 对用户输入文本进行过滤，以防止有害或不适当的文本指令被模型处理，以及 (2) 对上传的用户图像使用 NSFW Filter 来检测并阻止不适宜工作环境的内容 (NSFW) 或其他任何可能有害的图像输入。

幻觉。 类似于 LLMs，LLaVA 可能会产生与事实或输入数据不一致的输出。这尤其在关键应用（如医学）中引发了关于推理结果的担忧。

偏差。 偏差可以从基模型传递到 LLaVA，既来自视觉编码器 (CLIP)，也来自语言解码器 (LLaMA/Vicuna)。这可能导致偏差的结果或对多样化内容的不公平表示。

能源消耗。 尽管由于预训练数据集较小（详见 Sec. C），能源消耗不是 LLaVA 的主要问题，但在扩大预训练数据集或增加模型规模（例如扩展到更大的 LLaMA 版本，如 65B 模型）时，这可能成为一个问题。

评估复杂性。 评估 LLaVA 的表现具有挑战性，因为它涉及语言和视觉任务。我们的评估基准涵盖了多个方面，包括准确率、概念覆盖、推理能力和创造力。然而，还需要考虑其他方面，例如视觉内容幻觉的程度以及对视觉内容的细粒度理解。虽然文本-only 的 GPT-4 基于多模态评估在我们的研究中是一致且准确的，但在不同情况下的鲁棒性和评估其他未探索方面的能力是未来的工作课题。

尽管存在这些风险，我们认为向研究社区发布 LLaVA 的好处大于潜在的危害。它允许对模型进行持续的研究和改进，并促使社区参与开发更好的缓解策略来解决这些担忧。此外，LLaVA 的发布可以推动新应用和研究方向的发展，最终有助于推进基础模型在视觉-语言任务中的进步和负责任的部署。

B 更多结果

我们展示了 LLaVA 更多的定性结果，以分析其涌现行为和观察到的弱点。有关 LLaVA 在学术基准上的更多定量结果，请参考带有视觉指令微调的改进基线 [32]。在表 9 中，LLaVA 展示了与其论文中的另一个示例中 GPT-4 相似的行为。类似于 OpenAI 的 GPT-4 实时演示，LLaVA 能够基于图 2 中的简化用户输入草图生成用于交互式笑话网站的 HTML/JS/CSS 代码，尽管存在一个小错误。如图 3 所示，LLaVA 可以以对话风格遵循用户的指示并提供详细的响应或创意写作。此外，LLaVA 能够将视觉内容与预训练 LLM 中的文本知识联系起来，如图 4 和图 5 所示。

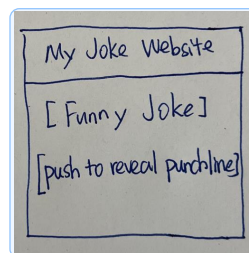
一个有趣的现象是，LLaVA 能够理解训练数据中未涵盖的视觉内容。例如，在图 6 中，它能够识别出 Elon Musk，无论是头像还是以狗造型出现的幽默表情包，尽管在 LLaVA 的视觉特征对齐或视觉指令微调阶段的训练数据中，Elon Musk 从未出现过。此外，LLaVA 在

表 9 和图 2 中展示了令人印象深刻的 OCR（光学字符识别）能力，而这些内容在我们的训练数据中很少涉及。

我们希望这些附加的结果和观察展示了 LLaVA 在各个应用领域的潜力。在未来的工作中，深入研究这些涌现的行为以及理解潜在的机制以使 LLaVA 表现出如此广泛的泛化能力是重要的。这将为构建更好的 LMMs 铺平道路，包括增强鲁棒性、减少偏差、以及改进学成的视觉-语言表示的对齐和范围。

Visual input example, Chicken Nugget Map:	
Sometimes I just look at pictures of the earth from space and I marvel at how beautiful it all is. 	
User	Can you explain this meme in detail?
LLaVA	The meme in the image is a creative and humorous take on food, with a focus on chicken nuggets, as the center of the universe. The meme begins with the phrase "Sometimes I just look at pictures of the Earth from space and I marvel at how beautiful it all is..." The main image features a tray of cooked chicken nuggets, with the chicken pieces positioned to resemble the continents and islands on a world map. The meme playfully suggests that the chicken nuggets represent the Earth, and the various locations depicted in the photo are actually chicken nugget versions of different places. The punchline of the meme appears in smaller text at the bottom, reading "I mean, it's not the real Earth, but how beautiful it is all is." This lighthearted and entertaining meme combines food and humor, using the image of chicken nuggets to create a funny and imaginative representation of the world.
User	Can you explain this meme?
GPT-4 [36]	This meme is a joke that combines two unrelated things: pictures of the earth from space and chicken nuggets. The text of the meme suggests that the image below is a beautiful picture of the earth from space. However, the image is actually of chicken nuggets arranged to vaguely resemble a map of the world. The humor in this meme comes from the unexpected juxtaposition of the text and the image. The text sets up an expectation of a majestic image of the earth, but the image is actually something mundane and silly.
User	Can you explain this meme in detail?
BLIP-2	sometimes i just look at pictures of the earth from space and marvel how beautiful it is
User	Can you explain this meme in detail?
OpenFlamingo	It's a picture of a chicken nugget on the International Space Station.

表 9: Example prompt comparing LLaVA, GPT-4, BLIP-2, and OpenFlamingo’s visual reasoning capabilities in understanding the humor. BLIP-2 and OpenFlamingo fail to follow the user’s instructions. LLaVA and GPT-4 both explain the meme and its humor, while GPT-4 produces a more concise answer.

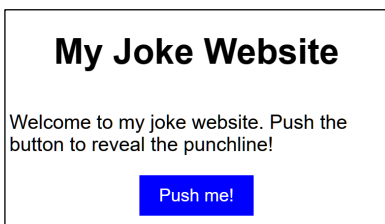


User

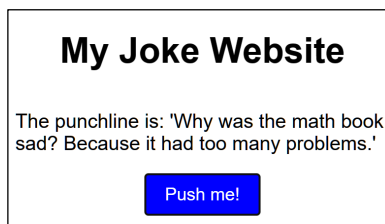
Turn this mock-up into an interactive website using html/js, replace the joke with a real joke.

```
<!DOCTYPE html>
<html>
<head>
<title>My Joke Website</title>
<style>
body { background-color: white; font-family: Arial, sans-serif; }
.container { display: flex; flex-direction: column; align-items: center; justify-content: center; height: 100vh; }
h1 { font-size: 36px; margin-bottom: 20px; }
p { font-size: 20px; }
button { background-color: blue; color: white; border: none; padding: 10px 20px; font-size: 18px; cursor: pointer; }
</style>
</head>
<body>
<div class="container">
<h1>My Joke Website</h1>
<p id="punchline">Welcome to my joke website. Push the button to reveal the punchline!</p>
<button onclick="revealPunchline()">Push me!</button>
</div>
<script>
function revealPunchline() {
document.getElementById("punchline").innerHTML = "The punchline is: 'Why was the math book sad? Because it had too many
problems.'";
}
</script>
</body>
</html>
```

Place the generated HTML code into an HTML renderer to render the web page.



Rendered website



Rendered website
(clicked the button)

图 2: LLaVA 根据用户草图输入生成用于交互式网站的 HTML/JS 代码。在修复生成输出中的一个小错误（红色部分）后，交互界面可以正常工作。LLaVA 的输出仍有改进空间，例如将笑话和笑点分成两行，并在点击按钮时才显示笑点，以更好地反映用户的意图。

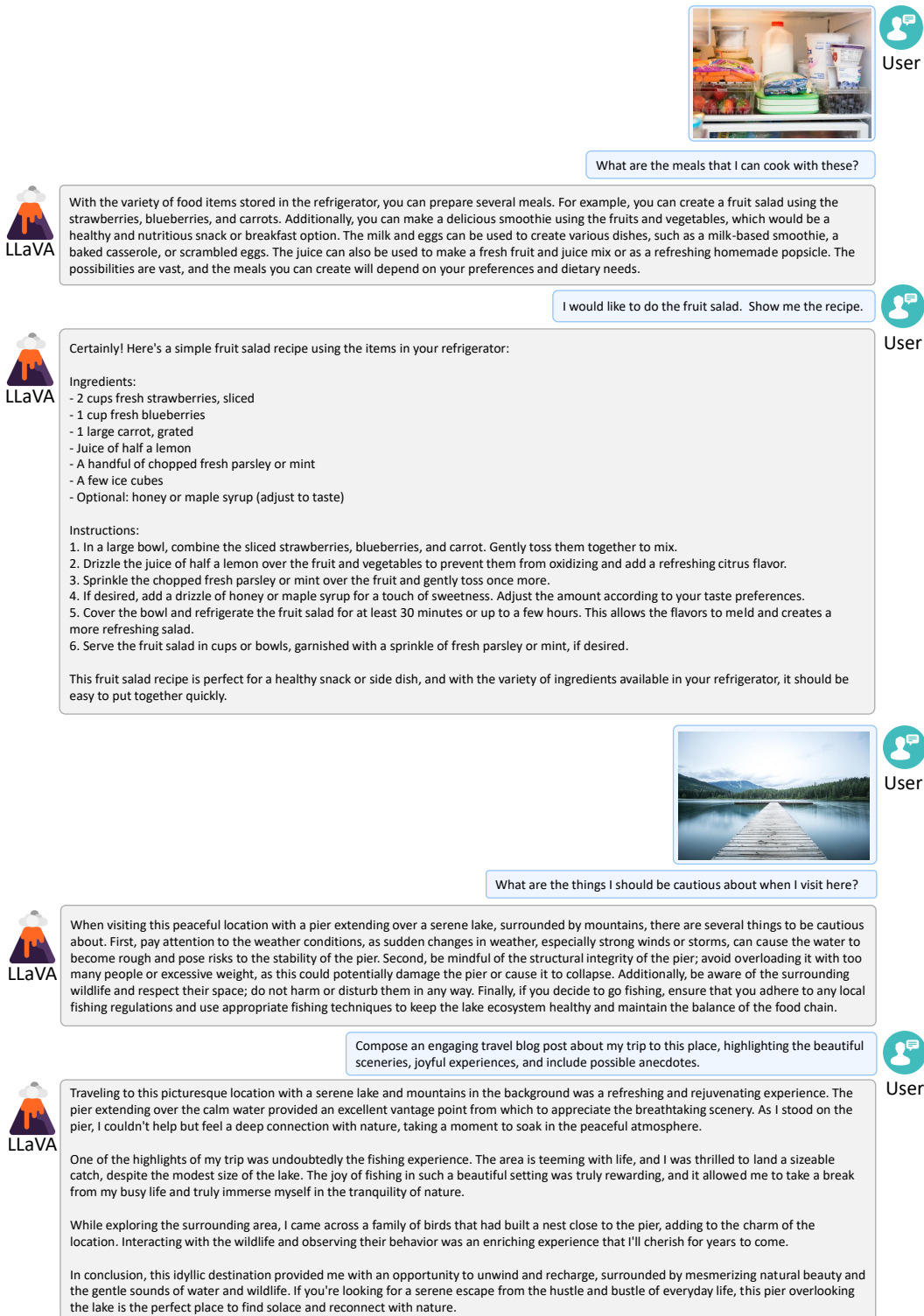


图 3: LLaVA 能够识别人体视觉内容, 而无需直接提示视觉识别。当接收到后续请求时, 它还会提供详细的响应, 生成的响应与提供的视觉内容密切相关。

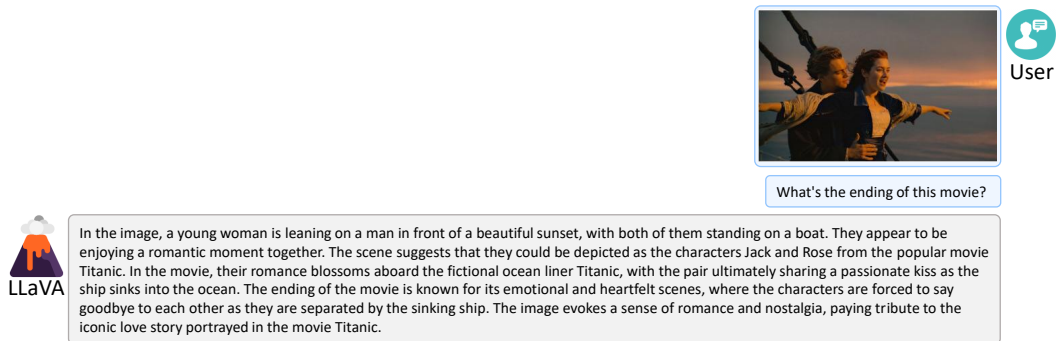


图 4: LLaVA 将电影场景与来自预训练 LLM 的文本知识相关联。

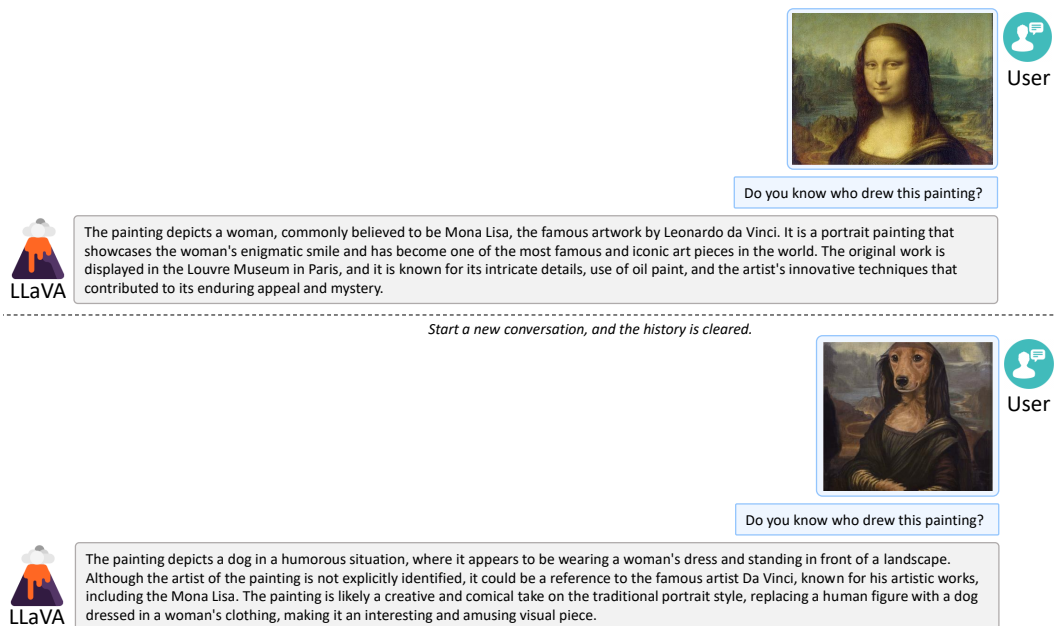


图 5: LLaVA 认出了达芬奇的著名艺术作品《蒙娜丽莎》。当我们开始一段新对话时，它还会解释在网络上模仿《蒙娜丽莎》创作的幽默艺术作品。

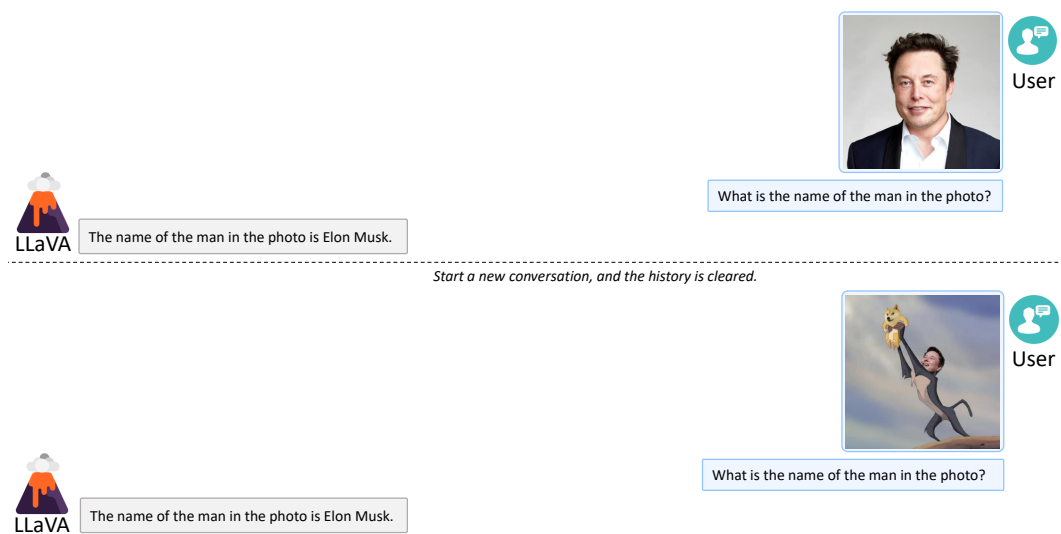


图 6: 一个有趣的涌现行为是, LLaVA 能够在头像图片和他穿着 doge 服装的搞笑表情包中都识别出埃隆·马斯克。这表明预训练的 CLIP 视觉编码器可能已经见过埃隆·马斯克的图像。然而, 这仍然令人惊讶, 因为无论是 LLaVA 的视觉特征对齐阶段还是视觉指令微调阶段的训练数据中, 埃隆·马斯克从未出现过, 这表明基础语言模型能够泛化到未见过的视觉概念。

Question:

Which material is this rocking chair made of?

Options: (A) wood (B) silk

LLaVA answer:

LECTURE: A material is a type of matter.

Wood, glass, metal, and plastic are common materials.

Some objects are made of just one material.

Most nails are made of metal.

Other objects are made of more than one material.

This hammer is made of metal and wood.

SOLUTION: Look at the picture of the rocking chair. The rocking chair is made of two different materials. The legs are made of wood, and the back and seat are made of silk.

The answer is B.

GPT-4 (text-only) answer:

The answer is A. BECAUSE: Rocking chairs are typically made of sturdy materials that can support the weight of a person and withstand the rocking motion. Wood is a common material used for making rocking chairs due to its strength, durability, and natural appearance.



GPT-4 (text-only) judge:

Since there is no context or image to refer to, I cannot independently analyze the rocking chair in question. However, I can try to analyze the reasoning given by the assistants.

Assistant 1, who claims to have seen the image, mentions that the legs of the rocking chair are made of wood, and the back and seat are made of silk, implying that it's a combination of two materials. However, this response seems less plausible, as silk is not a common material used for the construction of rocking chairs due to issues with stability and durability. Typically, one would expect the frame, legs, and seat or back to be made of wood, with cushions and padding made of fabrics, such as cotton or polyester.

Assistant 2 argues that rocking chairs are typically made of sturdy materials, and wood is a common material used for making rocking chairs due to its strength, durability, and natural appearance. This reasoning stands true and is based on common knowledge.

Given the information available, the most plausible answer is that the rocking chair is made of wood, as suggested by Assistant 1. The answer is A.

表 10: 一个样例展示了仅文本的 GPT-4 如何充当裁判, 集成 LLaVA 和一个仅文本的 GPT-4 的预测, 并给出正确的最终答案。

C 训练细节

我们使用过滤后的 CC-595K 子集对模型进行 1 轮预训练，学习率为 $2e-3$ ，批量大小为 128，并在提出的 LLaVA-Instruct-158K 数据集上进行 3 轮微调，学习率为 $2e-5$ ，批量大小为 32。遵循 Vicuna 的方法，我们使用 Adam 优化器，无权重衰减，以及带有 3% 预热比的余弦学习率。在微调过程中，使用 FSDP（全分片数据并行）和梯度检查点来节省 GPU 内存，未使用卸载。启用 BF16 和 TF32 以在速度和精度之间取得平衡。

我们使用 $8\times$ 个 A100 显卡训练所有模型。在 CC - 595K 上的预训练在 4 小时内完成。在 Instruct - 158K 上的微调在 10 小时内完成。在 ScienceQA 上的微调在 4 小时内完成。

D 资产

我们的源代码、生成的指令微调数据以及提出的基准已上传至匿名化的 GitHub 仓库：[LLaVA-Annonymous/LLaVA](#)。

1. 源代码: [link](#)
2. README: [link](#)
3. 启动演示的说明: [link](#)
4. 所有用于查询 GPT-4 的提示和少量样例: [link](#)
5. LLaVA-Instruct-158K: [link](#)
6. LLaVA-Bench: [COCO](#), [In-The-Wild](#)
7. 模型检查点。压缩后的模型检查点大小为 25GB，超过了 GitHub LFS（大型文件存储）的 5GB 限制。我们将把检查点发布给公众，或者在评审员要求时根据本次提交进行发布。

E 数据

简要图像描述的指令。 用于简要描述图像内容的指令列表如表 11 所示。它们用自然语言的变化表达了相同的含义。

详细图像描述的指令。 用于详细描述图像内容的指令列表如表 12 所示。它们用自然语言的变异表达了相同的含义。

CC3M. 我们使用 Spacy 为整个 CC3M 数据集中的每个描述提取名词短语，并统计每个唯一名词短语的频率。我们跳过频率小于 3 的名词短语，因为它们通常是其他描述已经涵盖的概念和属性的罕见组合。然后我们从剩余频率最低的名词短语开始，将包含该名词短语的描述添加到候选池中。如果名词短语的频率大于 100，我们随机从其所有描述中选择一个大小为 100 的子集。这最终产生了大约 595K 个图像-文本对。

过滤前后的名词短语统计量的比较如图 7 所示。过滤后的数据集显示了概念的良好覆盖，其频率高于 3，但图像-文本对的数量较少。

F 提示词

用于从 ChatGPT/GPT-4 生成基于图像的对话的提示如表 13 所示。

- "Describe the image concisely."
- "Provide a brief description of the given image."
- "Offer a succinct explanation of the picture presented."
- "Summarize the visual content of the image."
- "Give a short and clear explanation of the subsequent image."
- "Share a concise interpretation of the image provided."
- "Present a compact description of the photo's key features."
- "Relay a brief, clear account of the picture shown."
- "Render a clear and concise summary of the photo."
- "Write a terse but informative summary of the picture."
- "Create a compact narrative representing the image presented."

表 11: 简要图像描述的指令列表。

- "Describe the following image in detail"
- "Provide a detailed description of the given image"
- "Give an elaborate explanation of the image you see"
- "Share a comprehensive rundown of the presented image"
- "Offer a thorough analysis of the image"
- "Explain the various aspects of the image before you"
- "Clarify the contents of the displayed image with great detail"
- "Characterize the image using a well-detailed description"
- "Break down the elements of the image in a detailed manner"
- "Walk through the important details of the image"
- "Portray the image with a rich, descriptive narrative"
- "Narrate the contents of the image with precision"
- "Analyze the image in a comprehensive and detailed manner"
- "Illustrate the image through a descriptive explanation"
- "Examine the image closely and share its details"
- "Write an exhaustive depiction of the given image"

表 12: 详细图像描述的指令列表。

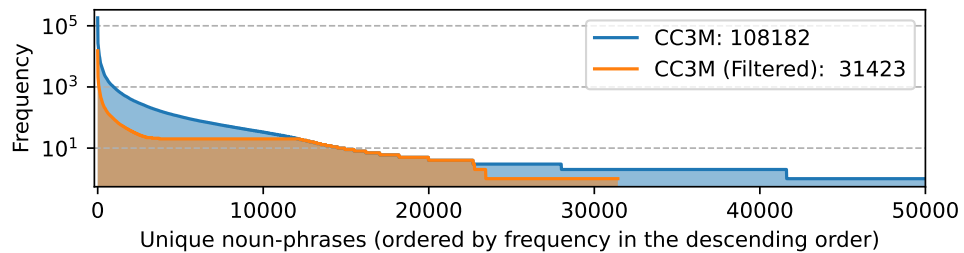


图 7: 比较过滤 CC3M 之前后的名词短语统计量。图例中报告了名词短语的独特总数。

```
messages = [ {"role": "system", "content": f"""You are an AI visual assistant, and
you are seeing a single image. What you see are provided with five sentences, describing
the same image you are looking at. Answer all questions as you are seeing the image.
```

Design a conversation between you and a person asking about this photo. The answers should be in a tone that a visual AI assistant is seeing the image and answering the question. Ask diverse questions and give corresponding answers.

Include questions asking about the visual content of the image, including the object types, counting the objects, object actions, object locations, relative positions between objects, etc. Only include questions that have definite answers:

- (1) one can see the content in the image that the question asks about and can answer confidently;
- (2) one can determine confidently from the image that it is not in the image. Do not ask any question that cannot be answered confidently.

Also include complex questions that are relevant to the content in the image, for example, asking about background knowledge of the objects in the image, asking to discuss about events happening in the image, etc. Again, do not ask about uncertain details. Provide detailed answers when answering complex questions. For example, give detailed examples or reasoning steps to make the content more convincing and well-organized. You can include multiple paragraphs if necessary."""}

```
]
for sample in fewshot_samples:
    messages.append({"role": "user", "content": sample['context']})
    messages.append({"role": "assistant", "content": sample['response']})
messages.append({"role": "user", "content": '\n'.join(query)})
```

表 13: 对于每个查询, 我们展示通过 ChatGPT/GPT-4 使用 few-shot 上下文学习方法从 `query['context']` 收集 `query['response']` 的提示构造过程, 其中示例来自 `fewshot_samples`, 每个示例包含输入 `sample['context']` 和输出 `sample['response']`。请注意, `messages` 是最终提示。在这个例子中, 我们提供了生成对话响应所用的提示, 请参见表 15 和表 16 了解其上下文学习示例的详细信息。我们建议读者查看代码库以获取生成其他两种类型响应 (包括详细描述和复杂推理) 的提示。

Context type 1: Captions

A group of people standing outside of a black vehicle with various luggage

Luggage surrounds a vehicle in an underground parking area

People try to fit all of their luggage in an SUV.

The sport utility vehicle is parked in the public garage, being packed for a trip.

Some people with luggage near a van that is transporting it.



Context type 2: Boxes

person: [0.681, 0.242, 0.774, 0.694], person: [0.63, 0.222, 0.686, 0.516], person: [0.444, 0.233, 0.487, 0.34], backpack: [0.384, 0.696, 0.485, 0.914], backpack: [0.755, 0.413, 0.846, 0.692], suitcase: [0.758, 0.413, 0.845, 0.69], suitcase: [0.1, 0.497, 0.173, 0.579], bicycle: [0.282, 0.363, 0.327, 0.442], car: [0.786, 0.25, 0.848, 0.322], car: [0.783, 0.27, 0.827, 0.335], car: [0.86, 0.254, 0.891, 0.3], car: [0.261, 0.101, 0.787, 0.626]

Response type 1: conversation

Question: What type of vehicle is featured in the image?

Answer: The image features a black sport utility vehicle (SUV).

Question: Where is the vehicle parked?

Answer: The vehicle is parked in an underground parking area, likely in a public garage.

Question: What are the people in the image doing?

Answer: The people are trying to fit all of their luggage into the SUV, likely preparing for a trip.

Response type 2: detailed description

The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip.

In the parking area, various luggage items are scattered around the vehicle. There are two backpacks, one located near the left rear wheel and the other closer to the right side of the vehicle. Additionally, there are two suitcases, one on the right side of the car and another further away near the center of the parking area. A bicycle can also be seen on the left side of the vehicle.

Other cars are parked around the main SUV, with one car positioned behind it and slightly to the left, another behind and slightly to the right, and the third car further behind on the right side.

Response type 3: complex reasoning

Question: What challenges do these people face?

Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings to accommodate. They might have to strategize and arrange the luggage efficiently to ensure that everything fits properly into the vehicle. Additionally, they need to consider the comfort of the passengers and visibility while driving, so the placement of the luggage must not obstruct the driver's view or make the passengers uncomfortable during the trip.

表 14: 一个说明指令跟随数据的示例。顶部的块显示用于提示 GPT 的上下文，例如标题和框，底部的块显示三种类型的响应。请注意，视觉图像未用于提示 GPT，我们仅在此处将其作为参考展示。

Captions

There is a movie theater that displays the show times above the doors.

A red fire hydrant is deep in the snow.

The fire hydrant is in the snow near a recently plowed sidewalk.

This city has had a very hard winter with snow.

A hotel for dogs in the snow in winter.

Conversation

Question:

What color is the fire hydrant in the image?

====

Answer:

The fire hydrant in the image is red.

====

Question:

Is the fire hydrant fully covered by snow or partially visible?

====

Answer:

The fire hydrant is partially visible, with a significant portion of it covered by the deep snow.

====

Question:

What is the condition of the sidewalk near the fire hydrant?

====

Answer:

The sidewalk near the fire hydrant has been recently plowed, so it is relatively clear of snow compared to the surrounding area.

====

Question:

What does the image suggest about the winter season in the city?

====

Answer:

The image suggests that the winter season in the city has been quite harsh and challenging. The significant snow accumulation seen in the image indicates that the city has experienced heavy snowfall, which can lead to various difficulties and disruptions for its residents.

For instance, the deep snow covering the fire hydrant could potentially hinder its accessibility in case of emergencies, which poses safety concerns. The recently plowed sidewalk implies that the city's maintenance crews have been actively working to keep the walkways clear and safe for pedestrians, but the sheer amount of snowfall might make it difficult to keep up with the continuous snow removal efforts.

Furthermore, such extreme winter conditions can impact transportation, leading to traffic issues, delays in public transit, and increased risks of accidents due to slippery roads. It can also cause problems for businesses and schools, as people might face challenges commuting to work or attending classes. Additionally, the heavy snow can put extra strain on infrastructure, such as roofs and power lines, increasing the likelihood of structural damage or power outages.

In conclusion, the image of the red fire hydrant deep in the snow and the recently plowed sidewalk suggest that the city has faced a particularly severe winter season, with substantial snowfall that has likely caused various challenges and disruptions for its residents and infrastructure.

表 15: 在即时学习中用于构建视觉对话数据的一个示例。

Captions

A man is skiing in the open snow covered hills

A skier is making his way into the snow.

A skier on the foothills of a huge mountain range.

A skier looks at mountains while standing near a trail sign.

a single person skiing by an area with a lot of bushes

Conversation

Question:

What is the skier doing?

===

Answer:

The skier is skiing in the snow-covered hills, making his way through the snow and enjoying the beautiful mountain scenery. They are also standing near a trail sign, which indicates their location on the mountain.

===

Question:

Are there any other people or skiers visible in the image?

===

Answer:

No, there is only a single person skiing in the image, and they are surrounded by an area with a lot of bushes.

===

Question:

What kind of skiing is the person doing? Cross-country or downhill?

===

Answer:

Based on what I can see in the photo, it is not possible to definitively determine whether the skier is doing cross-country or downhill skiing. However, given the presence of a trail sign and the fact that they are skiing in the foothills, it is more likely that the skier is engaged in cross-country skiing. Downhill skiing typically takes place on more steep slopes and involves ski lifts, whereas cross-country skiing often occurs on relatively flatter terrain with marked trails.

===

Question:

Can you explain cross-country and downhill skiing a bit more?

===

Answer:

Cross-country skiing and downhill skiing are two distinct types of skiing, each with its own techniques, equipment, and terrain preferences.

1. Cross-Country Skiing:

Also known as Nordic skiing, cross-country skiing involves traversing relatively flat terrain or gentle slopes using skis and poles. It is a popular recreational and competitive sport that focuses on endurance, balance, and aerobic fitness. Cross-country skiers use their own locomotion to move across snow-covered terrain, rather than relying on ski lifts or other forms of assistance.

There are two main techniques in cross-country skiing:

- Classic: This technique involves skiing in parallel tracks, using a forward-and-backward gliding motion while keeping the skis parallel to each other. The skier uses their poles to propel themselves forward.
- Skate skiing: This technique is similar to ice skating, where the skier pushes off one ski at an angle and glides on the other. It requires a smooth, groomed surface and is typically faster than the classic technique.

Cross-country ski equipment consists of lightweight skis, boots, bindings, and poles. The boots are more flexible than those used in downhill skiing, allowing for better ankle movement and control.

2. Downhill Skiing:

Also known as alpine skiing, downhill skiing involves descending slopes at high speeds using skis and poles for balance and control. This sport is more focused on speed, technique, and navigating challenging terrain, including steep slopes, moguls, and even jumps.

Downhill skiing can be further categorized into several disciplines, such as slalom, giant slalom, super-G, and downhill racing. Each discipline has its own set of rules, courses, and ski equipment.

Downhill ski equipment includes heavier and stiffer skis, boots, bindings, and poles than those used in cross-country skiing. The boots are more rigid to provide better support and control during high-speed descents and sharp turns.

In summary, cross-country skiing is an endurance-based sport that involves traveling across flat or gently sloping terrain, while downhill skiing is focused on speed and technique as skiers navigate steeper slopes and challenging terrain. Both sports require specialized equipment and techniques, but they offer different experiences and challenges to participants.